

Post-quantum PKI migration for federal agencies to meet NIST CNSA 2.0 deadlines and protect citizen data, addressing the complexity of large-scale certificate authorities.

Public / Government sector × Post-quantum cryptography migration × Post-quantum cryptography · OS017 v3 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
71 is the world moving	61 can we play, can we win	MEDIUM 0 named in the evidence	€429k observed evidence · per year	NEXT regulation adopted or propose...	L0 20 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

Federal agencies face a multi-year, multi-domain migration to post-quantum cryptography (PQC) to meet NIST CNSA 2.0 deadlines, protecting citizen data from harvest-now-decrypt-later attacks. This opportunity is for public-sector organizations with large, complex certificate authorities that need to transition to quantum-safe algorithms. The urgency is driven by finalized NIST standards and regulatory deadlines, yet enterprise adoption remains critically low, creating a pressing need for expert guidance.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€429k €429k – €939k	€15k €15k – €33k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

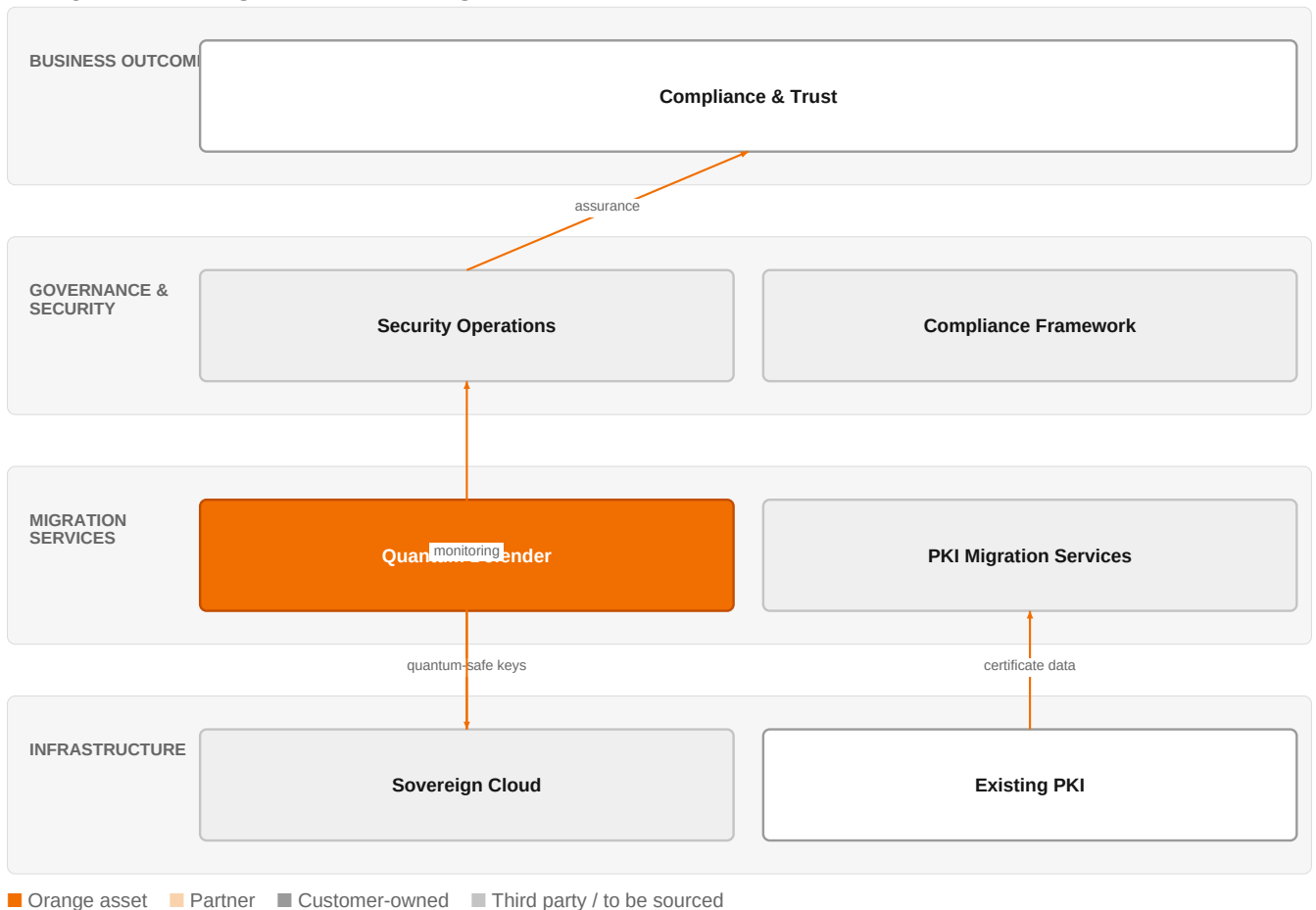
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Post-Quantum PKI Migration for Federal Agencies



This diagram shows how Orange's quantum-safe offers and sovereign infrastructure enable a secure, compliant migration of existing PKI systems.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

NIST has finalized post-quantum cryptography standards (FIPS 203, 204, 205), initiating an unprecedented cryptographic migration, yet enterprise adoption remains critically low. Regulatory deadlines are set just a few years in the future, and adversaries can already record encrypted traffic to decrypt later. PKI migration is particularly complex because these systems serve millions of users and are fundamental to cybersecurity, requiring coordination across libraries, protocols, and key stores.

Sources: SIG-B5F1E5C72CAB Springer Science and B 2026-07-30; SIG-C822609B6A07 openalex.org 2026-05-31; SIG-B8561992126B openalex.org 2026-07-30; SIG-444983B4C996 openalex.org 2026-05-09

Who buys, and why now

The Chief Information Security Officer (CISO) or Chief Technology Officer (CTO) of a federal agency signs the contract, but the pain is felt by the PKI operations team responsible for managing certificate authorities. The trigger is the approaching CNSA 2.0 deadline for new national security systems, forcing agencies to inventory their cryptographic assets and plan migration. The operational pain includes the complexity of re-issuing certificates across millions of users, ensuring interoperability, and avoiding service disruption.

Why it is hot now — every claim bound to a dated source

- Large-scale quantum computers threaten the public-key cryptography that protects enterprise data, communications, and digital identity. [SIG-40B64E273856, SIG-B8561992126B]
- Adversaries can record encrypted traffic today and decrypt it later, a tactic known as harvest-now-decrypt-later. [SIG-B8561992126B]

- As relevant guidelines are setting the deadlines for this migration just a few years in the future, it is due time to plan for effective mitigation and migration plans. [SIG-C822609B6A07]
- Federal post-quantum cryptography migration is scoped around three categories of cryptographic assets: libraries, protocols, and key stores. [SIG-444983B4C996]
- The migration of Public Key Infrastructures poses specific challenges, as these systems are complex, serving millions of users and fundamental in providing cybersecurity. [SIG-C822609B6A07]
- Relevant guidelines are setting the deadlines for this migration just a few years in the future. [SIG-C822609B6A07]
- The migration of Public Key Infrastructures poses specific challenges, as these systems are complex, serving millions of users. [SIG-C822609B6A07]
- Institutional approaches to post-quantum cryptography show a comparative analysis of migration frameworks from NIST, ANSSI, and BSI. [SIG-353070727CBB]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a team of Cybersecurity experts and Defense and security experts to conduct a cryptographic asset inventory, identify libraries, protocols, and key stores, and develop a migration roadmap. Leveraging Orange Group researchers, we would design a hybrid certificate strategy and implement quantum-safe algorithms using our Quantum Defender / quantum-safe offers. The engagement would follow a phased approach: assessment, planning, pilot, and full migration, with continuous monitoring and compliance validation against ISO 27001 and SecNumCloud standards.

Why Orange can win

Orange brings a unique combination of sovereign, certified infrastructure (SecNumCloud, CISPE) and deep expertise in defense and security, which is critical for federal agencies. Our Quantum Defender / quantum-safe offers are specifically designed for this transition, and our reference with Grand Paris Sud demonstrates our ability to handle large-scale public-sector projects. However, our assets are relatively thin compared to specialized competitors, so we must emphasize our trust and sovereignty credentials.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.4; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include global systems integrators like Atos/Eviden, IBM, and Accenture, who have broad PQC offerings and strong public-sector presence. Security specialists such as Thales, Airbus Protect, Entrust, and ID Quantique offer focused cryptographic solutions. The customer will compare our approach against these players, who may have more mature PQC tooling or deeper vertical expertise. However, our sovereign and certified infrastructure is a differentiator that many competitors lack.

Sources: SIG-353070727CBB openalex.org 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; active in Public / Government sector; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; active in Public / Government sector; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; active in Public / Government sector; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Public / Government sector; competes in Cybersecurity	structural	—
Accenture	Global systems integrator	active in Public / Government sector	structural	—
Capgemini	Global systems integrator	active in Public / Government sector	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you inventoried all cryptographic assets across your agency, including libraries, protocols, and key stores?
- 2 What is your current timeline for meeting CNSA 2.0 requirements, and have you identified any critical dependencies?
- 3 How many certificate authorities do you operate, and what is the volume of certificates you manage?
- 4 Have you experienced any harvest-now-decrypt-later concerns for sensitive citizen data?
- 5 Do you have a governance framework in place for coordinating PQC migration across different departments?

Objections you will meet

They will say	You can answer
We already have a plan with a major integrator.	That's a valid approach, but our sovereign, certified infrastructure and specialized quantum-safe offers can complement your existing plan, especially for the most sensitive systems.
PQC migration is too far out; we have more pressing priorities.	The deadlines are only a few years away, and harvest-now-decrypt-later means data is at risk today. Starting now reduces the risk of a rushed, error-prone migration later.
We don't have the budget for a full migration.	We can phase the migration to align with budget cycles, starting with a pilot on critical systems. Our approach is designed to minimize disruption and optimize costs.

Risks and unknowns

The migration is a multi-year, multi-domain transformation that could stall due to the sheer complexity of PKI systems and the need for ecosystem coordination. There is uncertainty about the exact timeline for quantum computers becoming a real threat, and the effectiveness of hybrid approaches is still being validated. Additionally, the discovery of cryptographic material in unexpected places, such as machine learning model parameters, could expand the scope beyond current inventories.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a federal agency's post-quantum PKI migration, differentiating Orange's approach by highlighting our quantum-safe offers, our expertise from cybersecurity and defense experts, and our reference with Grand Paris Sud, while ensuring alignment with SecNumCloud and CISPE certifications.

Sales	In a meeting with a federal agency's chief information security officer, say: 'Given the NIST deadlines for post-quantum migration, we should discuss how Orange's Quantum Defender and quantum-safe offers can help you address the complexity of large-scale certificate authorities, backed by our ISO 27001 and SecNumCloud certifications.'
Strategist / Innovator	Commission a deep-dive brief on the comparative migration frameworks from NIST, ANSSI, and BSI, focusing on how Orange's quantum-safe offers can be positioned for federal PKI migration, leveraging our cybersecurity and defense experts and Orange Group researchers.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B5F1E5C72CAB	2026-07-30	Springer Science and Busi...	1	Post-Quantum Cryptography Migration Readiness in Global Capability Centres: A Governance Framew...
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-57F4F0DAE044	2026-06-18	Cryptography	1	Mythos-Class Frontier Models and the Compression of Post-Quantum Cryptography Migration Timelin...
SIG-C822609B6A07	2026-05-31	openalex.org	1	Large Scale Actualisation of Post-Quantum Cryptography Migration
SIG-F119CA024DC5	2026-05-27	openalex.org	1	Post-Quantum Cryptographic Migration Strategies for Enterprise Systems
SIG-444983B4C996	2026-05-09	openalex.org	1	Parameter-Resident Cryptographic Material as an Unscoped Surface for Post-Quantum Migration: An...
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-969C5FDB15DD	2026-02-02	Institute of Electrical a...	1	Post-Quantum Cryptography Migration Framework for Real-Time Payment Gateways
SIG-53F3508BD517	2026-01-09	openalex.org	1	Post-Quantum Public Key Infrastructures: Hybrid Certificates, Cryptographic Combiners, and Migr...
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-B1382A6ECA9C	2025-12-22	openalex.org	1	pqc_migration_emulator.py — Professional Post-Quantum Cryptography Migration Emulator
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-0C3636BF137F	2026-05-27	bing.com	2	The Cryptographic Migration Clock Just Got Real: A Small-Cap Just Released The Tooling Stack Fo...
SIG-DD1DB909E321	2026-08-03	arxiv.org	3	Measuring Post-Quantum TLS Deployment Across UK Internet Sectors
SIG-2FE78254EE94	2026-07-31	arxiv.org	3	Mind the Gap: Policy vs Reality in Post-Quantum TLS Deployment
SIG-47DFEDD3D7A2	2026-07-15	arxiv.org	3	Multivariate Cryptography-Based Anonymous Certificate Scheme
SIG-69DE97B1F613	2026-06-05	arxiv.org	3	Empirical Evaluation of Large Language Models for Migration of Code Fragments to Post-Quantum C...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (4 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:41:14+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.